

**Transportation Access and Employment Outcomes Across ZIP Codes in Maricopa County,
Arizona Final Report**

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Introduction and Rationale

Transportation is very important for creating economic opportunities because it affects people's ability to get to work, school, and other important services. Reliable transportation systems help workers get to job centers quickly, but limited access can keep people from working. As metropolitan areas expand geographically, disparities in transportation access often emerge, particularly in regions where public transit coverage is uneven, or commute distances are increasing. Because of this, transportation has become an important focus area for applied data analytics, allowing researchers and policymakers to examine how infrastructure and mobility patterns intersect with economic outcomes.

Maricopa County, Arizona, provides a particularly compelling context for examining transportation access and employment outcomes. As one of the fastest-growing counties in the United States, Maricopa County has experienced rapid population growth alongside expanding suburban development. This growth has made the distance between homes and major job centers greater, making commute time and transit availability even more important. Some parts of the county have public transportation systems like buses and light rail lines, but other parts still rely heavily on cars. These differences make it easier or harder for people in different parts of the county to find work.

This subject is personally significant as transportation is a system that directly influences daily existence and living standards. Even without a clearly defined career path, analyzing transportation data provides an opportunity to apply programming and analytical skills to a real-world problem with visible impacts. Through coursework in data analytics and programming, this project allows for the application of data cleaning, regression analysis, and geographic

interpretation while exploring a domain that has relevance across multiple industries, including public policy, urban planning, and business analytics.

The broader significance of this topic is supported by existing research linking transportation access to economic stability and community well-being. Public sector studies in Maricopa County have demonstrated that expanded transit access can improve access to employment and other essential services, particularly in transit-dependent communities (Maricopa County Department of Public Health, 2017). This project builds on previous research by focusing on employment outcomes and using a data-driven approach to better understand how transportation characteristics affect labor market indicators at the county level.

Research Objective and Questions

The main goal of this proposed data analysis project is to look at how access to transportation affects job outcomes in Maricopa County, Arizona. The project aims to investigate whether differences in transportation accessibility, including how long it takes to get to work and how often public transportation is available, are linked to differences in unemployment rates and labor force participation.

The proposed research questions guiding this analysis are as follows:

1. Is greater access to transportation associated with lower unemployment rates in Maricopa County?
2. Do areas with longer average commute times experience poorer employment outcomes?

The project will mainly use regression analysis to answer these questions. Unemployment rate and labor force participation are two examples of employment indicators that are continuous dependent variables. This means that regression modeling is a good way to analyze them. The

project will probably have one main regression model, but there may be more exploratory models depending on initial findings and instructor feedback.

Preliminary Data Sources and Scope

This analysis will mainly use PolicyMap as its source for data. PolicyMap provides access to a wide range of demographic, economic, and transportation-related indicators aggregated at multiple geographic levels. This project will focus on Maricopa County as the geographic unit of analysis, using county-level variables related to unemployment, labor force participation, commute time, income, and transportation characteristics. PolicyMap is great for looking at the complicated links between transportation and job outcomes because it can combine data from many public sources.

Along with PolicyMap, other regional data will be used to help with the analysis. The Maricopa Association of Governments (MAG) Workforce Commute Explorer provides information about how people in the area commute, how jobs flow, and workforce mobility within the region.

While this tool may not be used directly in regression modeling, it will help inform variable selection and support interpretation of observed patterns by illustrating how residents travel to and from employment centers.

Data from Maricopa County's Labor Market Trends resources can also assist in understanding the labor market at the county level. The reports give the most recent information about the county's job market, including the number of people who are unemployed, the types of jobs available, and the types of people who work there. Adding this information will help ensure sure that the analysis is in line with current economic conditions and observed labor market trends.

Finally, findings from the Maricopa County Department of Public Health's transportation and health impact assessment will be used to support the conceptual framework of the project. This assessment shows that expanding transportation in Maricopa County has made it easier for people to find jobs and economic opportunities, especially in areas where people don't have easy access to cars (Maricopa County Department of Public Health, 2017). Even though this study only looks at job outcomes, the results show that having access to transportation is still an important factor in economic well-being.

Feasibility and Next Steps

Because the data sources that were found are easy to access and trustworthy, this project is thought to be very feasible. PolicyMap and county-level resources are readily available through the Mesa Community College library system, ensuring consistent access throughout the project timeline. By only looking at Maricopa County, the analysis is focused while still having a diversity of data to compare. The technical skills learned in the course match the analytical methods needed, such as data cleaning, exploratory analysis, and regression modeling.

There may be problems with data aggregation at the county level and the inability to establish direct causation between transportation access and employment outcomes. These problems will be addressed by carefully interpreting the results and being open about the limitations. The project will focus on correlation and association, not on causal claims.

Based on instructor feedback, the next steps will be to choose the final variables, get the necessary data from PolicyMap, and do exploratory data analysis to find trends and outliers. Next, regression models will be made and tested. The results will be looked at considering the state of transportation and the job market in the area. The last step will be to put together the

results into a clear story that shows how access to transportation affects job outcomes in Maricopa County.

Literature Review

Transportation access is widely acknowledged as a structural factor influencing economic opportunity. Having reliable access to transportation affects a person's ability to get to work, keep a regular work schedule, and be part of the workforce. In big cities with lots of different neighborhoods, differences in transportation infrastructure can make it harder for people to get to work in some neighborhoods than others. Because of this, more people are using data-driven methods to look at transportation systems and see how they affect the economy and society.

The Spatial Mismatch Hypothesis discussed by Kane suggests that employment opportunities and residential locations are often geographically disconnected, particularly for lower-income populations (Kane, 1992). Unemployment and underemployment may rise when workers reside far from employment hubs and lack dependable transportation. Later studies have confirmed this link by showing that people who rely on public transportation and cars have better job prospects (Blumenberg & Ong, 2001). These results suggest that transportation access may serve as both a direct and indirect predictor of labor market participation.

Another transportation-related factor that is often linked to economic outcomes is the time it takes to get to work. Longer commute times are linked to more stress, less job security, and less access to larger job markets (Han et al., 2022). In many cities, especially those with growing suburbs, lower-income families often have longer commutes because they must use public transportation or work far away (Blumenberg & Pierce, 2014). This means that the length of a person's commute may be a sign of structural transportation problems.

Transportation infrastructure is also important for the economy, according to public policy research. The Centers for Disease Control and Prevention (CDC) says that expanding public transportation systems is an evidence-based way to improve access to jobs and other important

services within five years (Maricopa County Department of Public Health, n.d.). A health impact assessment of light-rail expansion in Maricopa County found that better access to public transportation made it easier for people to find jobs, especially in neighborhoods where people didn't own cars (Arizona Department of Health Services, 2015). This regional evidence bolsters the theoretical link between transportation accessibility and economic engagement.

Even though there is a lot of research at the national and metropolitan levels, there are fewer studies that look at differences within a single rapidly growing county using smaller geographic units of analysis. Maricopa County is a unique case because its population is growing quickly, its suburbs are spreading out, and different neighborhoods have different access to public transportation. Examining transportation and employment relationships at the ZIP code (ZCTA) level in Maricopa County facilitates a more accurate analysis of localized variation, as opposed to depending on aggregated county-level averages. This project wants to add to the body of research by using regression analysis on ZIP code-level data to find out if access to transportation is a good predictor of unemployment rates and labor force participation in Maricopa County.

Hypothesis Development

The literature and conceptual framework discussed above will inform the hypotheses that will direct the data analysis for this Capstone project. All the hypotheses will be tested using ZIP code (ZCTA)-level data from PolicyMap that comes from Maricopa County.

Hypothesis 1: Commute Time and Unemployment

Research Question: Across ZIP codes (ZCTAs) within Maricopa County, is average commute time associated with unemployment rates?

Null Hypothesis (H_0), Two-Tailed Test: Average commute time has no statistically significant effect on unemployment rates in Maricopa County.

Alternative Hypothesis (H_1), Two-Tailed Test: Average commute time has a statistically significant effect on unemployment rates across ZIP codes in Maricopa County.

Model: Linear regression model with unemployment rate as the dependent variable and average commute time as the independent variable, measured at the ZIP code (ZCTA) level within Maricopa County using PolicyMap data.

Rationale: ZIP code–level variation allows for more precise testing of whether longer commute times function as a barrier to employment access within the county.

Hypothesis 2: Transportation Access and Labor Force Participation

Research Question: Across ZIP codes within Maricopa County, does greater transportation access increase labor force participation rates?

Null Hypothesis (H_0), One-Tailed Test: Transportation access has no statistically significant effect on labor force participation rates across ZIP codes in Maricopa County.

Alternative Hypothesis (H_1), One-Tailed Test: Greater transportation access is associated with higher labor force participation rates across ZIP codes in Maricopa County.

Model: Linear regression model with labor force participation rate as the dependent variable and transportation access indicators (e.g., average commute time, percentage of households with vehicle access) as independent variables at the ZIP code (ZCTA) level.

Rationale: If transportation improves access to workplaces, then ZIP codes with stronger transportation access should demonstrate higher workforce participation.

Hypothesis 3: Transportation Access and Median Household Income

Research Question: Across ZIP codes within Maricopa County, is transportation access related to median household income?

Null Hypothesis (H_0), Two-Tailed Test: Transportation access has no statistically significant relationship with median household income across ZIP codes in Maricopa County.

Alternative Hypothesis (H_1), Two-Tailed Test: Transportation access has a statistically significant relationship with median household income across ZIP codes in Maricopa County.

Model: Linear regression model with median household income as the dependent variable and transportation access measures as independent variables at the ZIP code (ZCTA) level.

Rationale: Transportation infrastructure may influence economic opportunity and residential patterns, making it important to examine income differences across ZIP codes.

Hypothesis 4: Income as a Moderating Variable

Research Question: Across ZIP codes within Maricopa County, does median household income moderate the relationship between transportation access and unemployment?

Null Hypothesis (H_0), Two-Tailed Test: Median household income does not moderate the relationship between transportation access and unemployment rates across ZIP codes in Maricopa County.

Alternative Hypothesis (H_1), Two-Tailed Test: Median household income significantly moderates the relationship between transportation access and unemployment rates across ZIP codes in Maricopa County.

Model: Multiple regression model with unemployment rate as the dependent variable and independent variables including transportation access, median household income, and an interaction term (Transportation Access $\times$ Income) measured at the ZIP code (ZCTA) level.

Rationale: Income levels may alter how transportation barriers are experienced, potentially strengthening or weakening the association between transportation access and unemployment.

Data Collection Strategy

The goal of this Capstone project's data collection plan is to find out how access to transportation affects job outcomes in different ZIP codes (ZCTAs) in Maricopa County, Arizona. The primary data source for this study will be PolicyMap, accessed through the Mesa Community College Library. PolicyMap collects data from trusted sources, like the U.S. Census Bureau's American Community Survey (ACS), which makes it a reliable and complete place to find information on demographics, the economy, and transportation. PolicyMap is a good tool to use because it lets you compare ZIP codes across several different variables in the same system. This is important for looking at how transportation access and job outcomes vary from place to place.

The geographic scope of this study will concentrate on ZIP codes within Maricopa County, as this level of analysis offers adequate granularity to identify neighborhood-level disparities that may not be apparent at the county level. The time frame will use data from 2020 to 2024, which will show the most up-to-date and relevant socioeconomic and transportation trends. Using this time frame makes sure that the analysis includes current commuting habits and job market conditions while keeping the data sets consistent. The combination of geographic and temporal scope helps with the main goal of finding significant differences in transportation access and job outcomes within a single metropolitan area.

This study will utilize specific variables from PolicyMap's transportation data categories to operationalize transportation access. These include the average time it takes to get to work, which shows how much time commuting takes, and the mode of transportation to work, which shows how much people depend on public transportation compared to their own cars. Other factors, like the number of cars per household, will show how easy it is to get around, while the distance to the nearest public transit stop and the frequency of transit service will show how

accessible and adequate the public transportation system is. All these factors together give a multidimensional picture of transportation access, showing both how close they are and how useful they are.

The dependent variables in this study will include unemployment rate and labor force participation rate, both of which are continuous measures of employment outcomes. In order to account for broader socioeconomic influences, control variables such as median household income and population density will also be included. These variables are supported by prior research as key factors influencing employment patterns and will help isolate the specific impact of transportation access within the regression models.

Before the analysis, the dataset will go through a few steps to clean and prepare it so that it is consistent and reliable. This process will involve finding and fixing missing values, making sure that variable formats are the same, and combining datasets by using ZIP code identifiers as a common key. In some cases, variables may need to be normalized so that they can be compared across different scales of measurement. These steps are necessary to get the data ready for regression analysis and make sure that the results are both correct and easy to understand.

Modeling Strategy

Multiple linear regression will be the main modeling method for this Capstone project because the dependent variables, the unemployment rate and the labor force participation rate, are both continuous. Regression analysis is appropriate because it allows for the estimation of relationships between multiple independent variables and a single outcome variable while considering other factors that might affect the results. This method is good for looking at how transportation access and socioeconomic conditions together affect employment outcomes because it is both statistically sound and easy to understand.

The regression models will incorporate transportation-related variables, including average commute time, mode of transportation, vehicle availability, and indicators of public transit accessibility as primary predictors. To make sure that the effects of transportation access aren't mixed up with larger economic or demographic factors, socioeconomic factors like median household income and population density will be used as control variables. By adding these variables, the model will be able to better separate the link between access to transportation and job outcomes.

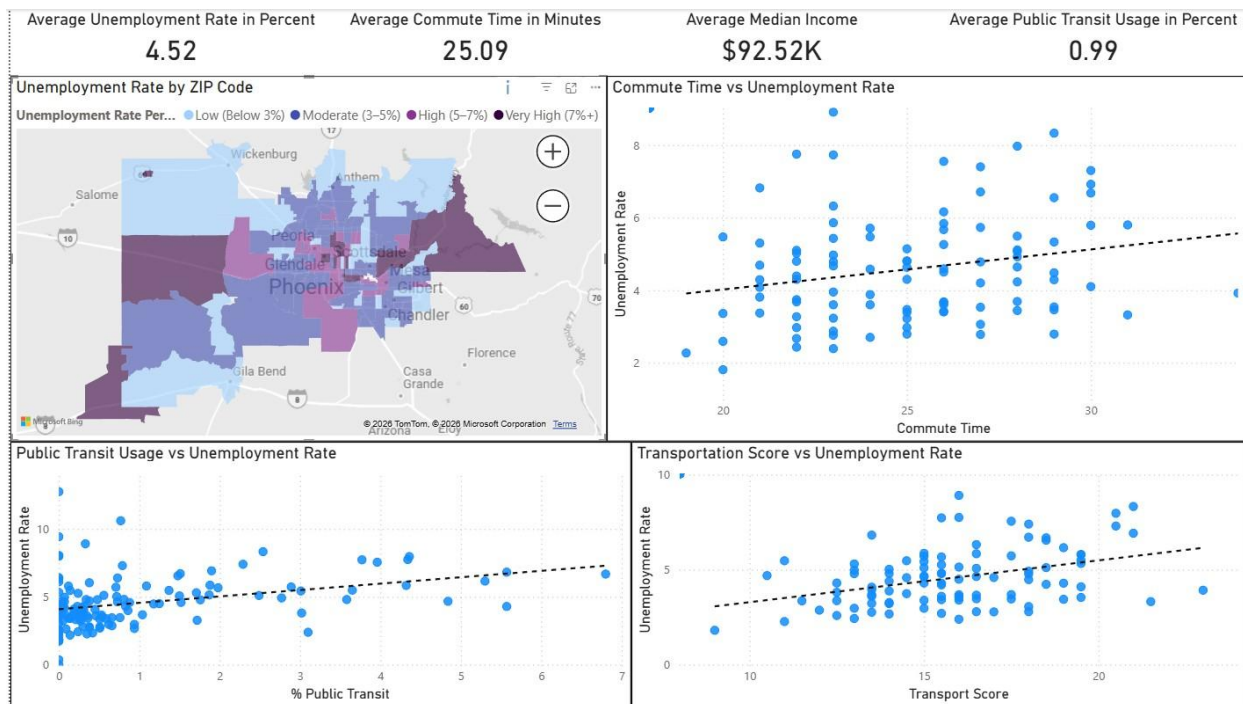
In addition to the primary regression model, an interaction term between transportation access and median household income may be included to test whether the impact of transportation varies across income levels. This is particularly important because transportation barriers may have a more significant effect on lower-income populations who have fewer mobility options. Adding interaction effects helps the analysis find more complicated relationships that might not be clear in a regular regression model.

The modeling process will follow a structured workflow consistent with best practices in data analysis. As outlined in Chapter 10, this includes data preparation, model development, model evaluation, and model selection (Suresh Kumar Mukhiya & Ahmed, 2020). Statistical tools like R^2 to measure explanatory power and p-values to measure statistical significance will be used to judge how well the model works. Also, the main assumptions of linear regression, such as linearity, independence, and homoscedasticity, will be checked to make sure the results are correct. These steps are necessary to make sure that the model gives useful and accurate information.

Overall, multiple linear regression is the most appropriate method for this analysis because it allows for the quantification of relationships between transportation access and employment

outcomes while controlling relevant socioeconomic variables. This method is directly in line with the research goal of finding out if access to transportation is a major factor in determining economic opportunity across ZIP codes in Maricopa County.

Dashboard Design and Visualization Analysis



This chapter describes how to design, build, and analyze a Microsoft Power BI dashboard that was made to look at the link between transportation access and job outcomes in Maricopa County, Arizona. The dashboard combines data on geography, income, and transportation at the ZIP code level to help with both exploratory and explanatory analysis. The main goal of the dashboard is to show patterns and relationships between important variables, such as the unemployment rate, the time it takes to get to work, the number of people who use public transportation, and a composite transportation score. The focus is on how each visualization relates to the research questions and hypotheses set out in earlier chapters, as well as how the design choices make it easier to understand and analyze.

The dashboard's overall layout is hierarchical, which means that it is meant to help the user go through the analysis in a logical order. A row of key performance indicators (KPIs) at the top of the dashboard gives a quick overview of the dataset. These KPIs are the average unemployment

rate, average commute time, average median income, and average public transit use in Maricopa County's ZIP codes. This summary layer helps users quickly understand the bigger picture of the economy and transportation before they look at more detailed charts. The dashboard is split into two main parts below the KPI row. The first part is a geographic visualization that shows spatial patterns in unemployment. The second part is a set of scatter plots that look at how transportation variables affect employment outcomes. This structure allows for both macro-level and micro-level analysis, which makes sure that the dashboard is a useful analytical tool instead of just a bunch of separate visuals.

The first graphic is a map that shows the unemployment rates by ZIP code. This visualization answers the research question of whether unemployment rates change from place to place in Maricopa County. The hypothesis that goes along with this is that unemployment rates are very different in different ZIP codes. This is a two-tailed test. A map visualization was chosen because the data is geographic by nature, and spatial representation makes it easy to see differences between regions right away. To make it easier to understand and cut down on the visual noise that comes with continuous gradients, unemployment rates are put into groups like low, moderate, high, and very high. The data structure uses ZIP codes to show where things are and categorized unemployment rates as the legend variable. The map shows that unemployment rates are higher in some western and peripheral areas than in others. In central and suburban areas, on the other hand, the rates are usually lower. These patterns suggest that structural factors that are specific to a certain area, such as transportation access, economic opportunity, and population distribution, can influence employment outcomes.

The second visualization uses a scatter plot with a linear trend line to look at the connection between the unemployment rate and the time it takes to get to work. This visualization looks at

the research question of whether longer commute times are linked to higher rates of unemployment. The null hypothesis posits that commute time does not influence unemployment, whereas the alternative hypothesis indicates the presence of a relationship. This is a two-tailed test because the direction of the relationship is not assumed ahead of time. A scatter plot was chosen as the best way to look at how two continuous variables are related to each other. The y-axis shows the unemployment rate, and the x-axis shows the average commute time in minutes. Adding a trend line lets you see how the relationship is going in general. The results show a small positive relationship, meaning that as commute times go up, so do unemployment rates. The data points, on the other hand, are spread out, which means that the relationship is not very strong. This implies that although commute times may influence employment outcomes, it is not a robust independent predictor and probably interacts with other socioeconomic variables.

The third graph looks at how the unemployment rate and the use of public transportation are related. You can also see a trend line on this scatter plot, which is used to see if using public transportation is linked to job outcomes. The corresponding hypothesis tests whether there is a statistically significant link between using public transportation and unemployment rates. The x-axis shows the percentage of people who use public transportation to get to work, and the y-axis shows the unemployment rate. The graph shows a positive trend, which means that more people using public transportation is linked to more people being unemployed. But the way the data points are spread out shows that they are mostly grouped together at low levels of transit use, which is a sign that people in Maricopa County don't use public transportation very much. This grouping shows that most ZIP codes rely heavily on private cars. The positive relationship observed is likely attributable to underlying socioeconomic conditions rather than a direct causal effect of transit usage itself. For instance, places with more people using public transportation

may also have more people living there or a lower median income, both of which can affect job outcomes. This interpretation shows how important it is to think about confounding variables when looking at transportation data.

The fourth and most important visualization looks at the link between a composite transportation score and the unemployment rate. This graph answers the question of whether looking at all the factors that affect transportation access together shows a stronger link with employment outcomes than looking at each factor on its own. The null hypothesis says that access to transportation doesn't affect unemployment, while the alternative hypothesis says that it does. This is also a test with two tails. The transportation score was based on several factors related to transportation, such as the average commute time, the number of vehicles available per household, and measures of how easy it is to use public transportation. To make sure that the scores could be compared, each part was normalized. The combined score was made so that higher values mean worse transportation access. This method gives a more complete picture of transportation barriers by measuring accessibility in more than one way instead of just one.

The x-axis shows the transportation score and the y-axis shows the unemployment rate. There is also a line that shows the trend. This graph shows a clearer and more consistent positive relationship than the last scatter plots. When the transportation score goes up, which means that transportation is less accessible, the unemployment rate also tends to go up. The points are closer together around the trend line, which means there is a stronger relationship than when looking at commute time or transit use alone. This finding supports the idea that transportation barriers, when looked at, have a significant effect on job outcomes. Using a composite score adds depth to the analysis by showing the combined effects of several transportation constraints. This makes it a stronger predictor in the model.

In all the scatter plot visualizations, there are a few outliers, especially when the unemployment rate is high. These outliers show ZIP codes where the unemployment rate is much higher than what would be expected based only on transportation variables. These outliers suggest that other things, like education levels, the types of industries in the area, or the state of the economy in that area, may be affecting job outcomes. These outliers shouldn't be seen as mistakes; instead, they add important context and show how complicated the link between transportation and employment is. Recognizing these deviations enhances the analysis by illustrating that transportation is merely one of several contributing factors rather than the sole determinant.

The dashboard also has interactive features that make it easier for users to get involved and analyze data. Users can hover over data points to see more information about each ZIP code, such as the exact values for the unemployment rate, commute time, and other factors. This feature lets you look more closely at localized patterns and helps you understand them better.

The scatter plots are easier to use because they all have the same design, with shared axes and formatting. This makes it easy to compare different transportation variables. Adding categorical grouping to the map visualization makes it easier for users to understand by breaking down complex data into clear ranges.

Summary of Hypotheses and Results

This study investigated the association between access to transportation and employment outcomes across ZIP codes in Maricopa County, Arizona. Several hypotheses were developed to test whether transportation-related variables including commute time, public transit usage, and a composite transportation score were associated with unemployment rates. The analysis results support the hypotheses to a different extent, which points to the complexity of the relationship and the necessity to have multiple factors in mind at the same time.

The first hypothesis tested for a significant relationship between commute time and unemployment rates. The null hypothesis is that there is no relationship between commute time and unemployment. The alternative hypothesis is that there is a relationship. Based on the scatter plot analysis and the observed trend line the null hypothesis was not rejected. The data showed a weak positive trend that suggested longer commute times could be correlated with higher unemployment rates, but the relationship was weak and the data points were scattered. This suggests that commute time alone is not a good predictor of employment outcomes, and that other variables are likely more important.

The second hypothesis investigated whether there is a relationship between public transit usage and unemployment rates. The null hypothesis predicted that there would be no relationship between public transit use and unemployment, while the alternative hypothesis predicted that there would be a relationship between the two. The results indicated a small positive trend, higher levels of transit usage being associated with higher unemployment rates. However, the distribution of the data showed heavy clustering at very low levels of transit usage, indicating a low reliance on public transit in the region. Because of the weak pattern and the potential for confounding socioeconomic factors, the null hypothesis was not rejected. We find that public transit usage does not meaningfully explain variation in unemployment and could instead be a

proxy for underlying demographic or economic conditions.

The third hypothesis explored if a composite transportation score is significantly related to unemployment rates. The null hypothesis was that transportation accessibility has no effect on unemployment, while the alternative hypothesis was that lower transportation access is associated with higher unemployment. The transportation score was created by combining normalized measures of commute time, vehicle availability, and use of public transit, with higher scores indicating worse transportation conditions. The scatter plot analysis showed a stronger and more stable positive correlation between transportation score and unemployment rate, with data points fitting better along the trend line than for the individual variables. This evidence led to rejection of the null hypothesis. Our findings suggest that transportation accessibility as a composite factor is significantly associated with employment outcomes.

In general, the results of the hypothesis tests indicate that individual transportation variables are weakly explanatory, but a more complete measure of transportation access provides stronger evidence of a relationship to unemployment. This underlines the importance of multi-dimensional analysis in solving complex socio-economic problems.

Project Summary and Reflection

The capstone project analyzed the relationship between transportation access and employment outcomes using Maricopa County ZIP code level data. Python was used for data cleaning, merging, and feature engineering techniques to create a comprehensive dataset that supports the analysis by combining datasets from PolicyMap. The project utilized Microsoft Power BI as the primary visualization tool for exploring statistical relationships and geographic patterns among key variables. Combining spatial and analytical visualizations allowed for a better understanding of the possible influence of transportation factors on unemployment rates.

One of the key findings of the project is that transportation accessibility does have a measurable effect on employment outcomes when considered as a combined factor. While individual variables like commute time and public transit use were weakly related to unemployment, the transportation composite score showed a stronger, more consistent pattern. This indicates that transportation barriers are complex and cannot be adequately captured by individual metrics. The geographic analysis supported these findings also, in that areas with higher unemployment rates were generally areas with less favorable transportation conditions.

Throughout the project several challenges were faced. One big challenge was the PolicyMap data, which required a lot of cleaning and restructuring due to inconsistent naming conventions and other formatting issues. Careful attention to detail was needed to select the right variables and to ensure alignment across datasets. In addition, creating a useful transportation score involved careful thought about how to combine and normalize different variables. Another challenge was to interpret findings that recognize both patterns and limitations, particularly in the presence of weak relationships and outliers.

Still, the project offered a lot of learning. It enhanced the skills of data cleaning, data integration and feature engineering with Python. It also enhanced Power BI skills, especially in developing dashboards that effectively present analytical insights. Most importantly, the project illustrated the need for critical thinking in data analysis, including the importance of questioning assumptions, recognizing limitations, and exploring alternative explanations. These lessons will directly apply to future work in data analytics and provide a strong foundation for more advanced analytical projects.

Real-World Implications

The results of this project have important real-world implications for transportation, employment, and socioeconomic inequality. Transportation access is vital to connecting people with job opportunities, and transportation limitations can create substantial barriers to employment. The results of this analysis suggest that transportation challenges are associated with higher rates of unemployment, indicating that transportation infrastructure and access are important determinants of economic participation.

In the case of Maricopa County, the low levels of public transit usage evident in the data suggest a strong dependence on personal vehicles. Such dependence can be a disadvantage for those who do not have access to a reliable vehicle, especially those in lower-income communities. Limited transportation access can restrict job search opportunities, increase commuting burdens, and reduce overall economic mobility. These challenges are particularly relevant in suburban and peripheral areas where job centers may be far from residential communities. The geographic patterns seen in the dashboard seem to point to transportation disparities as a factor in the uneven employment outcomes across the region.

From a public policy perspective, these findings highlight the importance of investment in transportation infrastructure and expanding access to reliable transit options. Reducing barriers to employment could include improving public transit systems, increasing service frequency, and enhancing connectivity between residential areas and employment centers. Also, policies that promote vehicle access, such as car ownership assistance programs or enhanced ride-sharing services, may contribute to solutions for transportation-related challenges. Understanding the connection between transportation and employment can help policymakers make more informed decisions that support economic inclusion and workforce participation.

The project's impact goes beyond the transportation aspect to address larger societal challenges like income inequality and opportunity access. The link between transportation and unemployment is closely associated with other factors including education, income, and housing. Limited transportation access can also be linked to challenges such as lower income levels or fewer economic resources, creating a cycle of disadvantage. Removing transportation barriers therefore can have a ripple effect that improves access to jobs, education and critical services.

Overall, this project adds to the understanding of the impact of transportation access on employment outcomes, and the importance of considering multiple factors in socioeconomic analysis. By integrating data-driven insights with practical considerations, the results lay the groundwork for future research and policy development to enhance transportation equity and economic opportunity.

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